3.1 Factoring $ax^2 + bx + c$

Name _____ Period ____ Date _____

Algebra II 3.1 · Student sheet

I can…

- factor out the GCF and factor $ax^2 + bx + c$.
- solve quadratic equations by factoring.
- find x-intercepts and solve real-life problems by factoring.

Vocabulary · fill in the missing word

Greatest common factor (GCF) — the greatest factor _____ by every term

Factoring — writing an _____ as a product

Zero-Product Property — if ab = _____, then a = 0 or b = 0

Zero of a function — an _____ where f(x) = 0; an x-intercept of the graph

Part 1 · Box it up

Complete each area box and write the factored form. Check by multiplying.

$x^2 + 7x + 12$	$2x^2 + 7x + 3$	$3x^2 - 10x - 8$

Part 2 · Notes and examples (fill in each note, then work the example)

Factor the GCF: Always factor out the _____ first.

Example 1

Factor out the GCF: $6x^2 - 15x$

Factoring $ax^2 + bx + c$: For $ax^2 + bx + c$, find factors of _____ that add to _____.

Example 2

Factor each expression.

a. $x^2 - 9x + 20$ b. $2x^2 + 5x - 12$

Zero-Product Property: Zero-Product Property: if ab = 0, then _____.

Example 3

Solve $3x^2 - 5x - 2 = 0$.

Example 4

A rectangular garden is 3 meters longer than it is wide, and its area is 40 square meters. Find its dimensions.

Part 3 · Partner practice

Partner 1

Factor $4x^2 - 4x - 3$.

Partner 2

Find the x-intercepts of $y = x^2 + x - 30$.